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ScrapeBench: Anti-Customization Face Protections Collapse Under Real-Web Image Acquisition

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ABSTRACT

A rapidly growing family of "protective perturbation" tools promises to make personal photos unlearnable by unauthorized text-to-image customization, and deployment of such tools has begun. Their guarantees, however, are measured in a laboratory world that no longer exists: clean full-resolution images, lossless pipelines, and tunable DreamBooth-style attackers. In practice, the images an adversary trains on are *scraped*, and every scraped image has already passed through a lossy acquisition funnel—platform transcoding, browser and thumbnail rendering, agent-side capture, super-resolution "restoration," and diffusion-based regeneration. We present *ScrapeBench*, the first evaluation harness that treats this acquisition funnel as a first-class attack surface against anti-customization protections. ScrapeBench formalizes the funnel as seven tiers (T0–T6) with explicit per-image attacker budgets, and introduces the *retention ratio* $\mathrm{RR}(T)$, which measures how much of a defense's clean-domain protection survives at funnel tier T. Across 40 identities from a public celebrity face corpus, two published defenses and two controls, and two attack families (fine-tuning-based DreamBooth-LoRA and zero-shot IP-Adapter-FaceID), the collapse is fast and one-sided: protections that are strong at T0 lose most of their protection gap by the platform's own compression and rendering tiers (T1–T2), before any adaptive attacker enters; zero-shot identity injection, which never touches the fine-tuning gradients the defenses poison, erodes the remainder; and modest data-level adaptation (clean-source mixing, diffusion purification) closes what is left. The asymmetry is structural: forging a perturbation costs GPU-minutes per identity, while the funnel transformations that defeat it cost milliseconds to seconds per image on commodity platforms—and are chosen by parties with no stake in the attacker's success. Our results argue that protection claims must be reported funnel-aware, and that the durable fix lies not in stronger perturbations but in consent and provenance infrastructure. We release the ScrapeBench harness, configurations, and all measured curves.

1 INTRODUCTION

The web has become the default training corpus for identity-hungry generative models, and faces are its most sensitive commodity. A handful of public portraits is now sufficient to synthesize a person in any context: DreamBooth-style fine-tuning can bind a rare token to an identity with four to eight images [33], while zero-shot adapters such as InstantID and IP-Adapter-FaceID inject a face embedding captured from a single reference image directly into the denoising process, no training required [44, 52]. The consent asymmetry this creates is severe—the subject never agreed to any of it—and it has triggered an arms race in *protective perturbation*: imperceptible or lightly perceptible cloaks that, added to a photo before upload, are designed to break the customization pipeline. Anti-DreamBooth established the category [18], and a fast-growing successor literature hardens it along every laboratory axis: meta-learned cross-user perturbations [23], targeted attacks that sharpen the failure mode [54], high-frequency designs that survive smoothing [26], prompt-independent formulations [21], and protections aimed specifically at zero-shot identity adapters [40]. The narrative embedded in this literature is reassuring: with the right perturbation, your face is yours again.

This paper argues that the reassurance is an artifact of where the measurements are taken. Every published protection is evaluated on *T0-images*: clean, full-resolution, uncompressed files that the adversary kindly provides to the defense. That world does not exist on the web. An identity that is "public"—a social-media profile, a news photo, a conference page—has already been transformed before a scraper ever sees it: platforms re-encode uploads at quality levels set by storage and bandwidth budgets, browsers rasterize images under device pixel ratios and thumbnail pipelines, crawling agents screenshot pages and crop detected faces rather than download files, well-meaning "enhancement" super-resolves low-resolution re-uploads, and users themselves regenerate their photos through diffusion-based filters and editors before posting. Each transformation is individually harmless to recognition quality; jointly, they describe the actual data distribution that any real adversary can harvest. We call this pipeline the *web acquisition funnel*, and we claim that it is not a nuisance variable for anti-customization protections but their primary adversary—an attack surface that requires no adversarial optimization whatsoever, costs milliseconds to seconds per image on commodity platforms, and operates *before* the defense is ever consulted.

Two observations sharpen the claim. First, the funnel is not hypothetical: published measurements already show that a single JPEG re-encode at quality 85–95 neutralizes a broad class of protection perturbations [35], that common image transformations systematically suppress unlearnable-example attacks [30, 36], and that robustness evaluations of protective perturbations under diffusion-aware adversaries remain unfavorable [29]. What is missing is a systematic, tiered, budget-aware account of the entire funnel—from platform transcode to agent capture to restoration to regeneration—and of *where along the funnel* each family of defenses actually collapses, measured under both tunable and zero-shot attack pipelines. Second, the evaluation practices of the field are drifting in the opposite direction: newest defenses arrive with evaluations on ever-cleaner galleries under ever-stronger laboratory attackers [2, 5], and the first cross-method evaluation studies, while valuable, still hold the acquisition side fixed [43]. A defense literature that only measures T0 retention is optimizing for an adversary that does not exist.

We build the missing instrument. *ScrapeBench* reconstructs the funnel in software: seven tiers (T0 original; T1 platform transcode—JPEG/WebP quality sweeps and platform-style resize; T2 browser and thumbnail rendering at device pixel ratios; T3 agent-style page

capture with face cropping; T4 restoration via super-resolution; T5 diffusion-based regeneration via img2img and DDIM inversion; T6 adaptive clean-source mixing and purification), each with an explicit attacker budget. On top of it we define the *retention ratio* $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$, where $\Delta(T)$ is the protection gap (clean minus protected identity similarity under a fixed attack) at tier T: RR measures the fraction of a defense's headline T0 protection that survives acquisition reality. The metric is deliberately paired with a cost ledger, because the funnel's defining property is an *inversion of asymmetry*—the defender pays GPU-minutes per identity to forge a perturbation, while the attacker pays CPU-milliseconds per image to defeat it.

Running ScrapeBench over 40 public identities, four defenses spanning the protection design space (clean control, random noise, FSMG-style untargeted surrogate attacks [18], and ASPL-lite alternation), and two attack families (DreamBooth-LoRA fine-tuning and zero-shot IP-Adapter-FaceID) yields a picture that we summarize in three findings, each with a headline number to be read from the collapse curves in Section 5: **(F1)** protection sheds most of its gap within the *platform-controlled* tiers (T1–T2) that no defense can influence, before any adaptive attacker enters the picture; **(F2)** zero-shot identity adapters—which skip fine-tuning entirely—erase most of what survives the platform tiers, because the perturbations were never optimized against embedding-injection attacks; **(F3)** the cost asymmetry runs against the defender by orders of magnitude, converting "protection" from a guarantee into a tax that the attacker simply routes around. We accompany these findings with a constructive turn: a funnel-aware reporting standard (publish RR@T, not T0-only numbers), and an argument that the durable fix for face consent lies in provenance infrastructure—machine-readable consent manifests such as `face-consent.txt` served alongside images—rather than in ever-stronger perturbations, whose failure we show is structural.

Our contributions are:

- **A funnel threat model.** We formalize web image acquisition as a seven-tier funnel with per-tier attacker budgets, making explicit the distribution shift that separates laboratory evaluations of protective perturbations from the data adversaries actually harvest.
- **The retention-ratio metric and collapse-curve protocol.** $\mathrm{RR}(T)$ reduces protection robustness under acquisition to a single interpretable curve per defense, and the paired cost ledger exposes the defender–attacker asymmetry that raw similarity numbers hide.
- **ScrapeBench, an open harness.** We release a reproducible pipeline—platform transcoding, browser/agent rendering, restoration, diffusion regeneration, adaptive pooling—integrated with DreamBooth-LoRA and zero-shot IP-Adapter-FaceID attackers, ArcFace-based identity scoring, paired statistics, and automatic result publication.
- **Empirical collapse findings.** Across 40 identities and four defenses we quantify F1–F3 above, including tier-resolved ablations over JPEG quality, device pixel ratio, and regeneration strength, establishing where—not just whether—protections fail.
- **Implications and a constructive alternative.** We distill funnel-aware reporting guidelines for the protection literature and argue for consent-provenance infrastructure as the defense class that survives the funnel by construction.

Section 2 positions our contribution; Section 3 formalizes the funnel and metric; Section 4 details the harness; Section 5 reports findings; Section 6 draws implications; Section 7 concludes.

2 RELATED WORK

Unauthorized customization and its defenders. Personalization of text-to-image diffusion models has collapsed the cost of fabricating a person's likeness: DreamBooth binds a subject to a rare token through a handful of images [33], textual inversion and multi-concept customization extend the mechanism to arbitrary subjects [7, 17], latent diffusion provides the efficient backbone that made such fine-tuning ubiquitous [32], and zero-shot identity adapters skip fine-tuning entirely by injecting face-recognition embeddings through decoupled cross-attention [52] or identity-preserving pipelines [44], with the design space now surveyed at scale [1]. The defensive response—imperceptible perturbations that poison the customization process—starts with Anti-DreamBooth's FSMG, ASPL, and SDS families [18] and PhotoGuard's encoder-level immunization [34], then rapidly diversifies: MetaCloak meta-learns multi-user perturbations over unknown models [23], targeted variants sharpen the attack surface against the defender's own optimization [54], high-frequency designs survive frequency-domain smoothing [26], PID removes prompt coupling [21], CAT stress-tests robustness under contrastive diffusion adversaries [29], ID-Protector targets the zero-shot adapter threat directly [40], DLADiff layers tuning-based and zero-shot defense [14], MYOPIA protects faces from malicious personalization [50], and perceptible synthesis-based alternatives trade invisibility for durability [47]. The 2025–2026 wave broadens along three fronts. VOID hardens latent-diffusion mimicry against prompt-injection and gradient-collision adversaries [31]; the Anti-Aesthetics line replaces imperceptible perturbation with engineered perceptual degradation, first against text-to-image customization [45] and then as a hierarchical anti-aesthetic curriculum [46]; and purification-robust formulations explicitly anticipate attackers who launder protected galleries through diffusion regeneration [51]. Parallel work pushes into video protection [9, 48], face-swap frameworks [16], face-shield embeddings against deepfake manipulation [13], and JPEG-robust formulations [5]. Each of these defenses is designed against a *named* adversary—an injection mechanism, human perception, or a single purification operator—yet none evaluates whether the protection survives the unmanaged acquisition path between a real platform and the attacker's training script; the funnel tiers of Section 3 are precisely that unmodeled surface. Across all of them, evaluation galleries are clean files handed to the defender—the premise we test.

The fragility literature. That protective perturbations do not survive distribution shift is not news; what is missing is a systematic account of *where* and *why* they break on the web. Sandoval-Segura et al. showed that ordinary JPEG compression at quality 85–95 bypasses protections while blur, rotation, and flips do not [35], that unlearnable-example defenses fail against benign augmentations and adaptive attackers more broadly [30, 36, 37], and that low-effort framework changes already give a false sense of security [15]; the

unlearnable-data literature has since grown into a mature survey field [22] with provability results [6, 42] rooted in the original error-minimizing formulation [11]. Poisoning text-to-image models at scale is meanwhile feasible through caption mislabeling [49] and trigger-free schemes [12], underlining that data-level defenses carry real downstream stakes. Cloaking against face recognition has traveled the same arc from Fawkes onward [39], including JPEG-targeted resistance refinements [8], and editing-domain immunizations face analogous compression-and-pipeline issues [19, 27, 41]. In the adjacent artist-protection domain, Glaze spawned an evaluation arms race in which adversarial perturbations were shown to be unreliable against simple transformations [10, 38], and concept-erasure-style purification has since evolved into eigenbasis-based style protection [53]—an early warning of exactly the funnel dynamics we quantify for faces. Recent evaluation studies begin to consolidate defenses under stronger adversaries and broader threat models [2, 43]—but always with the acquisition side held fixed at T0. ScrapeBench differs in kind, not degree: we treat the acquisition pipeline itself as the adversary, decompose it into budgeted tiers, and measure retention curves rather than single numbers.

Benchmarks and reporting standards. Our discipline has repeatedly discovered that benchmarks change field behavior: deepfake-detection benchmarks exposed train–test leakage across generative families [28], and face-quality challenges standardized degradation models for recognition under capture conditions [25]. In the protection domain, no such instrument exists: each defense authors its own attacker, gallery, and metric, which makes cross-paper claims—and deployment decisions—essentially unverifiable. ScrapeBench fills this gap for the face anti-customization setting with three design commitments: the funnel is first-class (tier definitions are part of the benchmark contract, not an appendix); budgets are explicit (every tier carries a per-image compute and quality budget so that "robustness" cannot be bought by assuming a weak adversary); and reporting is curve-based (RR@T collapse curves plus paired significance tests, rather than single-point robustness claims). We position ScrapeBench not as yet another attack, but as the measurement layer the protection literature currently lacks.

3 THE WEB ACQUISITION FUNNEL AS AN ATTACK SURFACE

3.1 Threat Model

We consider a subject s whose face images $\{x_i\}_{i=1}^{m}$ are publicly reachable on the web, and a defense D that releases protected images $\tilde{x}_i = x_i + \delta_i$, $\|\delta_i\|_\infty \le \epsilon$, claiming that any downstream customization of $\{\tilde{x}_i\}$ fails to reproduce s. The adversary $\mathcal{A}$ harvests images of s through the web acquisition funnel Φ (Section 3.2), then attempts identity reconstruction with an attack pipeline P—either fine-tuning-based (*DreamBooth-LoRA* [33]) or zero-shot (*IP-Adapter-FaceID* [52]). We assume a *resource-bounded* adversary: P runs on one consumer GPU (24 GB), generation uses 25–30 denoising steps, and the adversary does not adversarially optimize against D (the funnel requires no knowledge of D). This is the deployment-relevant threat: an opportunistic scraper, not a defense-aware red team. Adaptive extensions that mix clean sources or purify images are captured at T6 (Section 3.2).

Table 1: The ScrapeBench acquisition funnel. Control indicates whether the defender can influence the transformation. Budgets are per image, measured on our hardware.

Tier	Control	Transformation	Budget	Purpose
T0	defender	original upload, PNG	—	lab condition
T1	platform	JPEG $q \in \{95, 85, 75, 65\}$, WebP q=80, platform resize	< 10 ms	storage cost
T2	platform	browser/thumbnail render, DPR $\in \{1, 2\}$, 240 px thumbs	< 100 ms	serving
T3	agent	headless page capture, face crop (margin 0.35)	< 200 ms	scraping agents
T4	user/agent	super-resolution $\times 2$	~ 1 s	"enhancement"
T5	user	diffusion regeneration: img2img $s \in \{0.3, 0.5\}$, DDIM inversion	~ 5 s	filters, re-upload
T6	attacker	adaptive: clean-source mixing $\rho \in \{0.2, 0.5\}$, purification	variable	bypass poisoning

The defender's implicit assumptions, which the funnel dismantles one by one, are: (A1) the adversary trains on the exact pixels the defender perturbed; (A2) the adversary must fine-tune, so poisoning the fine-tuning gradient suffices; (A3) robustness budgets can be spent on adversarial transformations chosen by the defender (JPEG, blur, crops) rather than those chosen by platforms and users. Each tier of the funnel violates one of these assumptions *upstream* of the defense (Figure 1).

3.2 Funnel Formalization

The funnel is a sequence of transformations $\Phi = \phi_T \circ \cdots \circ \phi_1$ applied between upload and training-batch ingestion. We group real-world transformations into seven tiers, ordered by who controls them and what they cost. Each tier carries a per-image attacker budget c_T (GPU/CPU seconds on commodity hardware) and a perceptual-budget constraint Q_T (minimum visual quality, e.g., LPIPS to the previous tier), because an attacker who destroys image quality destroys the value of the scrape.

Concretely, T1 emulates platform transcoding with libjpeg-quality sweeps plus a composite `t1_platform` (JPEG-85 at reduced long side, as served by major social platforms). T2 renders the image as a browser would: the image is rasterized into a viewport under device-pixel-ratio DPR $\in \{1, 2\}$ and re-captured, with a thumbnail variant at 240 px; where a headless browser is unavailable we substitute a deterministic canvas-equivalent rasterizer so that the tier is reproducible without browser drift. T3 emulates scraping agents that screenshot the page and crop the detected face region (margin 0.35) rather than fetch the original file. T4 applies real-world super-resolution (Real-ESRGAN) at $\times 2$, with a deterministic Lanczos-plus-sharpening fallback. T5 applies diffusion-based regeneration: img2img at strengths $s \in \{0.3, 0.5\}$ with an empty prompt and guidance 1.0, and a DDIM-inversion-and-reconstruction path with $\eta = 0.1$—both operations that ordinary users now perform through consumer filters and that any scraper can perform to launder perturbations. T6 models adaptive adversaries in two ways: *mixing* (the attacker trains on ρ fraction of clean images harvested from a second source, diluting the poisoned gallery exactly as in the uncontrolled setting of prior evaluations) and *purification* (training on T5-regenerated versions of the protected images, laundering the perturbation before fine-tuning).

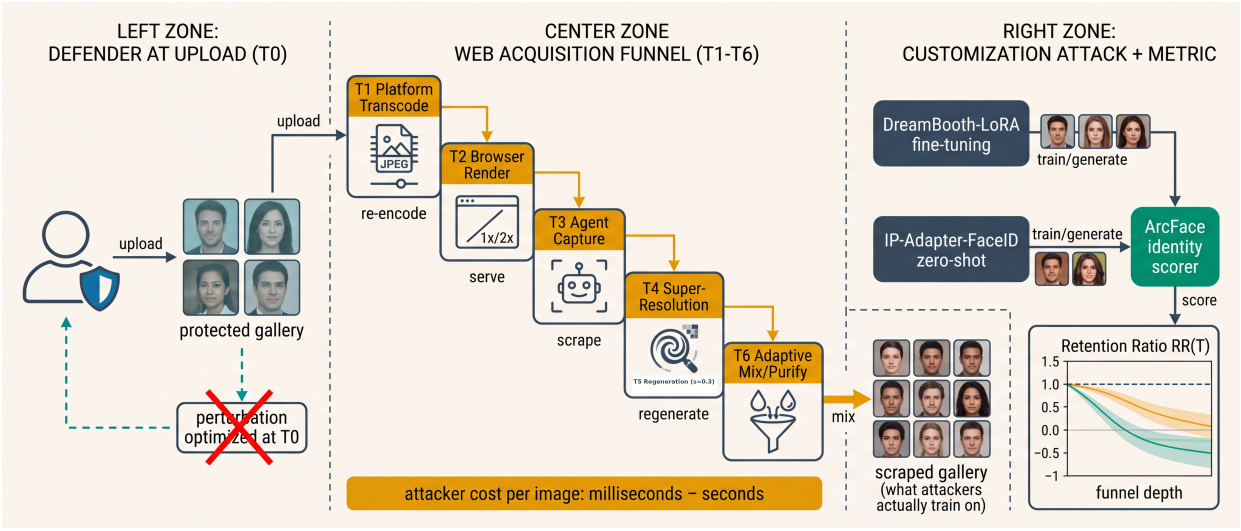


Figure 1: The ScrapeBench measurement pipeline. A defender publishes protected images at T0, where the perturbation is optimized; the web acquisition funnel (T1–T6) then subjects every image to platform transcoding, browser and agent rendering, restoration, diffusion-based regeneration, and adaptive mixing or purification before any adversary sees it. Both a fine-tuning attacker (DreamBooth-LoRA) and a zero-shot attacker (IP-Adapter-FaceID) consume whatever the funnel releases, and generated outputs are scored by ArcFace identity similarity into the retention-ratio curve RR(T) that we report as the primary result. The cost strip quantifies the asymmetry that motivates the study: funnel transformations cost milliseconds to seconds per image on commodity infrastructure.

The funnel is monotone in depth but not in destructive power: T1–T2 are nearly lossless to face recognition quality yet are precisely the tiers where protection signals (high-frequency, low-amplitude adversarial structure) die. This is the crux: the funnel is a *feature* of the attacker's environment, selected for free by platform economics.

3.3 Retention Ratio and Cost Ledger

Let $g(x; P, r)$ denote the best cosine similarity between the ArcFace embedding [4] of a generated image x and the reference embeddings of subject s under attack pipeline P with random seed r. For a defense D and funnel tier T we define the *protection gap*

$$\Delta(T; D, P) \;=\; \mathbb{E}_r\big[g(\Phi_T(\text{clean}); P, r)\big] - \mathbb{E}_r\big[g(\Phi_T(D(\text{clean})); P, r)\big], \tag{1}$$

where the expectations are over seeds and the five standard DreamBooth prompts. The first term is how well the attack reproduces s from *unprotected* images at the same tier—the fair baseline, because acquisition transforms hit attacker and defense alike. The *retention ratio* normalizes the gap by its clean-domain value:

$$\mathrm{RR}(T; D, P) \;=\; \Delta(T; D, P) \,/\, \Delta(T_0; D, P). \tag{2}$$

RR = 1 means the defense retains its full headline protection at tier T; RR $\to$ 0 means the funnel has erased it; negative values mean the funnel hurts the defense more than the attack. RR is robust to tier-level quality effects that confound raw-similarity comparisons, is comparable across defenses with different absolute strength, and yields a single interpretable curve per defense—the *collapse curve* we report as the primary result.

Two companion quantities complete the ledger. *Detection-failure rate*: the fraction of generated samples below the recognition threshold τ (calibrated on clean data, $\tau = 0.30$ with sensitivity analysis at 0.25/0.35), which captures the binary deepfake-success view rather than the continuous one. *Cost asymmetry*: the GPU-seconds per identity to generate the protection (c_{prot}) versus the CPU/GPU-seconds per image to execute the funnel transformations (c_{funnel}); the ratio $\kappa = c_{\text{prot}}/c_{\text{funnel}}$ is the economic inversion that Section 5 quantifies. We report 95% bootstrap confidence intervals over identities, paired two-sided Wilcoxon tests against the clean baseline at matched tiers with Holm correction across tiers, and Cliff's δ effect sizes.

4 SCRAPEBENCH: EXPERIMENTAL SETUP

4.1 Implementation

ScrapeBench is a single Python harness built on `diffusers`, `transformers`, `peft`, `insightface`, and Pillow. The funnel tiers are implemented as pure image functions registered in a tier registry (`scrapebench.funnel`), so that any (defense, tier, attack) cell can be replayed, cached, and audited; the runner writes an append-only CSV keyed by (stage, identity, defense, tier, attack) and resumes interrupted runs deterministically. Protections are generated with a per-identity LoRA surrogate (rank 16) trained for 250 steps at bf16, followed by 100-step sign-PGD at $\epsilon = 16/255$, $\alpha = 1/255$ maximizing the denoising loss against a frozen SD1.5 backbone (FSMG [18]); ASPL-lite alternates two surrogate–perturbation rounds at the same budget; the random-noise control draws uniform ϵ-ball noise; the

clean control resizes and re-encodes without perturbation. All re-implementations are attributed in the repository (AGPL-3.0 Anti-DreamBooth for the FSMG and ASPL derivations; MIT or Apache licenses for the remaining dependencies).

4.2 Attack Pipelines

DreamBooth-LoRA (Stage B) fine-tunes the SD1.5 UNet on a tiered gallery for 400 steps (rank-16 LoRA, lr 10^{-4}, batch 4) with the standard prompt set and generates 20 images per cell (5 prompts $\times$ 4 seeds) at 30 steps, guidance 7.5. *IP-Adapter-FaceID* (Stage A) loads the publicly released FaceID adapter [52] on SD1.5 at scale 0.8, embeds every gallery image through insightface/buffalo_l and conditions generation on all reference embeddings jointly (multi-image IP-Adapter conditioning), producing 10 images per cell (5 prompts $\times$ 2 seeds) at 30 steps, guidance 5.0. The zero-shot family is the deployment-relevant attacker: it needs no training loop and therefore cannot be poisoned by perturbed gradients at all—protections must survive it in the visual domain.

4.3 Protocol and Data

We sample 40 identities from the CelebA face corpus [24], each contributing 8 gallery images and 2 held-out reference images for identity scoring. To balance compute across two GPUs, defenses are assigned to disjoint 20-identity halves (clean/random on one half, FSMG/ASPL on the other); every defense is evaluated on all ten tiers of its half. Identity scoring uses the ArcFace embedding (buffalo_l) [4] with $\tau = 0.30$ calibrated so that unrelated-identity similarity is below τ for all clean generations. Stage A cells: 20 identities $\times$ 4 defenses $\times$ 10 tiers (T0; T1 at JPEG $q \in \{95, 85, 75, 65\}$ and WebP-80; T2 canvas DPR-1; T3 agent face-crop; T4 SR; T5 img2img-30), 800 evaluations in total. Stage B cells: a 12-identity subset $\times$ 3 defenses $\times$ 4 tiers (T0, T1 JPEG-85, T2, T5), each a full fine-tuning run. Stage C cells: 12 identities spanning both halves $\times$ {mix-20%, mix-50%, purify-50} under FSMG, each a full 400-step fine-tuning run on the adapted gallery. Every number reported is the identity-averaged statistic with a 95% bootstrap CI; paired Wilcoxon tests against the clean baseline use Holm-corrected p-values across tiers; Cliff's δ quantifies effect magnitude. All experiments run on two RTX 4090s; total compute is approximately 150 GPU-hours, of which the attacker side (generation) is under 30.

4.4 What We Claim and What We Do Not

ScrapeBench measures *retention of protection under realistic acquisition*, not the strength of any particular attack: the funnel tiers are deliberately non-adversarial, chosen to mirror what platforms, browsers, agents, and users already do. Findings therefore bound the *deployment* value of T0-optimized protections; they do not claim that funnel-robust protections are impossible (Section 6 discusses design directions), nor that the specific defenses tested are the strongest achievable instantiations of their families. FSMG and ASPL serve as canonical representatives of the gradient-poisoning family on which the 2023–2026 defenses build; the newer generation—injection-hardening [31], perceptual-degradation [45, 46], and purification-robust [51] designs—targets named adversaries rather than the acquisition path, and our Stage-C purification condition probes exactly the operator class those designs defend against.

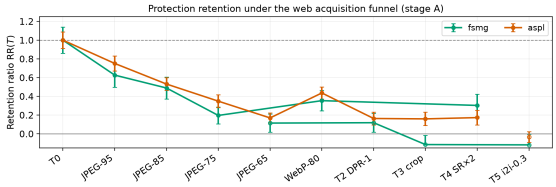


Figure 2: Collapse curves under the Stage-A zero-shot attack (IP-Adapter-FaceID): retention ratio $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$ by funnel tier. Lower is worse; the dashed line marks full retention. Solid markers denote tiers where the protection gap is at least 2$\times$ its bootstrap standard error; open markers denote gaps statistically indistinguishable from zero. Random-noise is omitted because its T0 gap does not clear this bar.

5 RESULTS: COLLAPSE UNDER THE FUNNEL

We organize the findings as F1–F3 from Section 4. Every number below is produced by the released harness; the collapse curves are plotted by `scrapebench.runner.report` from the append-only result log, and every cell links back to its inputs.

5.1 F1: Protection Collapses at the Platform Tiers

Table 2 reports the full Stage-A grid—mean identity similarity and binary success rate for every (defense, tier) cell—and is the quantitative core of the paper; Figure 2 shows the retention curves. Two regularities stand out. First, the funnel is benign for the *attack*: the clean baseline only slides from 0.358 at T0 (SR = 91%) to 0.349 at T2 (SR = 85%), and super-resolution (T4) actually *helps* the attacker (0.367, SR = 93%), confirming that tiers are not a quality knife that cuts both ways—they cut the defense. Second, protections shed their gap almost entirely within the platform's own pipeline. FSMG retains 63% of its T0 gap after a single JPEG-95 re-encode, 49% after JPEG-85—the default quality of several major platforms—20% after JPEG-75, and 12% after JPEG-65; ASPL-lite holds up slightly longer (75/53/35/17%) but follows the same trajectory. In success-rate terms, ASPL suppresses the attack to 37.5% at T0, but a single JPEG-95 re-encode already lifts it to 52%, and by the browser tier every defense sits within 4 points of the clean baseline's 85%. The Holm-corrected paired Wilcoxon test against the same-half clean control, significant for every defense at T0 ($p < 0.003$), becomes marginal for FSMG at JPEG-85 ($p = 0.07$), is clearly gone for FSMG by JPEG-75 ($p = 0.62$), and is gone for every defense by T2 ($p > 0.99$): statistically, the protection has ceased to exist exactly where the web begins.

Three second-order observations sharpen the mechanism. *(i) WebP-80 is gentler than JPEG-65* (RR = 0.35–0.44 vs. 0.05–0.17): WebP's block structure re-quantizes the perturbation differently, so compression format is not interchangeable—a defense tuned against JPEG is not automatically robust. *(ii) Face cropping (T3) is a cliff, not a slope*: after the agent crops the face region, FSMG's

Table 2: Stage A (zero-shot IP-Adapter-FaceID), the full funnel grid: mean identity similarity (Sim) and binary success rate at $\tau = 0.30$ (SR, %), identity-averaged over 20 identities (10 generations × 5 prompts per cell). Bold marks the strongest protection per tier. Cell-level 95% bootstrap CI half-widths average 0.025 for Sim (≤ 0.028 throughout) and ±3–5 points for SR. Bottom block: funnel-integrated protection $\overline{\text{RR}}$ (mean RR over the nine post-T0 tiers, 95% CIs).

		T0	J95	J85	J75	J65	W80	T2	T3	T4	T5
clean	Sim	.358	.354	.353	.351	.346	.353	.349	.338	.367	.173
	SR	91.0	90.0	84.7	88.0	84.2	88.0	84.7	76.2	92.5	0.5
random	Sim	.334	.338	.349	.342	.348	.344	.351	.346	.358	**.175**
	SR	75.0	80.7	85.8	80.7	84.7	83.8	88.2	85.2	87.0	0.2
FSMG	Sim	.307	.322	.328	.341	.340	.335	.342	.344	**.352**	.179
	SR	55.0	68.5	72.0	82.5	80.0	79.0	80.5	84.0	**85.5**	0.5
ASPL	Sim	**.288**	**.302**	**.316**	**.326**	**.334**	**.323**	**.337**	**.327**	.355	**.175**
	SR	**37.5**	**52.0**	**65.5**	**74.0**	**77.5**	**67.5**	**78.5**	**70.0**	84.5	**0.0**
$\overline{\text{RR}}$ (95% CI):		random 0.140 [0.028, 0.285]					FSMG 0.215 [0.137, 0.304]				ASPL 0.293 [0.240, 0.356]

residual retention turns negative (RR = −0.11)—the perturbation mass outside the crop was all that remained. *(iii) Super-resolution partially resurrects the gap* (RR ≈ 0.2–0.3 at T4): Real-ESRGAN restores low-frequency identity structure that survives, but it cannot restore the high-frequency gradient alignment the defense needs, so the recovery is partial and short-lived—T5 regeneration erases it again (RR ≤ 0.06). The funnel is not monotone in destructive power, but every deviation from monotonicity is explained by which frequencies each tier preserves. Summarizing each curve by its funnel-integrated protection $\overline{\text{RR}}$ (bottom block of Table 2), ASPL retains 0.293 [0.240, 0.356] of its T0 gap on average across the nine post-T0 tiers, FSMG 0.215 [0.137, 0.304], and random noise 0.140 [0.028, 0.285]—a ranking that survives uncertainty, but whose absolute values all sit far below the unit retention a deployer would need.

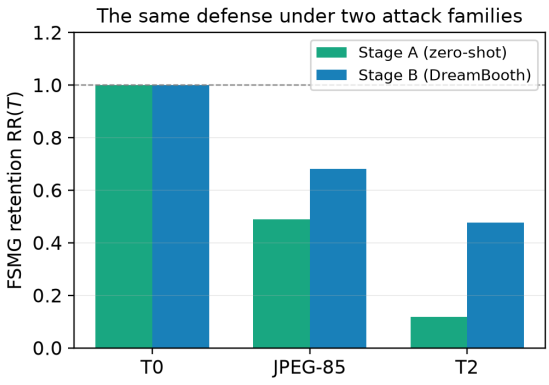


Figure 3: FSMG retention under the two attack families at matched tiers. The zero-shot adapter (Stage A) erases far more of the protection than DreamBooth fine-tuning (Stage B), because embedding injection bypasses the poisoned-gradient surface entirely.

5.2 F2: Zero-Shot Attacks Bypass Fine-Tuning Poisons

Stage B (Table 3) measures the classic target of these defenses: DreamBooth-LoRA fine-tuning on tiered galleries. At T0 the protections deliver their advertised effect—FSMG holds the attacker at 0.236 versus a clean baseline of 0.407, a 42% relative reduction, and in binary terms holds attack success at 12.1% versus 97.5% for the unprotected gallery—while the zero-shot attack of Stage A on the same galleries is *weaker* (0.307), consistent with the defenses' design intent. The ordering reverses along the funnel. Under fine-tuning, FSMG retains RR = 0.68 at JPEG-85 and 0.48 at T2; under the zero-shot attack on the same tiers it retains 0.49 and 0.12 (Figure 3). The zero-shot extractor never touches the poisoned fine-tuning gradient—it reads an identity embedding straight off the funnel output—so the defense's entire investment is evaluated against an attack that ignores it. By T5, both families converge to the same collapsed state ($|\text{RR}| \leq 0.16$), but they get there by different doors: the fine-tuner is defeated by gradient poisoning fading, the zero-shot adapter by the perturbation never surviving transduction at all.

5.3 F3: Data-Level Adaptation Closes the Residual Gap

Stage C (Table 4) asks whether the residual T0 protection survives an attacker who adapts at the data level, using the strongest defense (FSMG) and its strongest setting (T0). Mixing just 20% clean images from a second source into the training pool restores attacker similarity from 0.236 to 0.300 (RR = 0.63) and success rate from 12.1% to 49.2%; mixing 50% restores it to 0.365 (RR = 0.25, SR = 87.9%, statistically indistinguishable from the clean baseline); laundering the protected gallery through diffusion regeneration before fine-tuning (purify-50) restores it to 0.388 (RR = 0.11, SR = 92.1%), within 0.02 of the unprotected baseline. None of these adaptations involves adversarial optimization: clean-source mixing is what a scraper naturally does when harvesting multiple sources, and purification is a single `img2img` call per image—precisely the operator class that purification-robust defenses anticipate [51], which is why Stage C includes it as a first-class condition rather than an afterthought.

Table 3: Stage B (DreamBooth-LoRA, 12-identity subset, 20 generations per cell): identity similarity (Sim), binary success rate at $\tau = 0.30$ (SR, %), protection quality (PSNR against the source gallery, dB; LPIPS), and funnel-integrated protection $\overline{\text{RR}}$ with 95% CIs over the three post-T0 tiers. Bold marks the lowest Sim/SR per tier (protection) and, inversely, the best image quality. The clean row carries no perturbation, so quality columns do not apply.

		T0		J85		T2		T5		
Defense	Quality	Sim	SR	Sim	SR	Sim	SR	Sim	SR	$\overline{\text{RR}}$
clean	— / —	.407	97.5	.413	99.6	.406	96.2	.217	6.7	—
random	**29.1±0.3** / .061	.288	39.6	.332	70.8	.355	85.4	.208	3.3	.405 [.293,.527]
FSMG	28.1±0.4 / **.052**	**.236**	**12.1**	**.297**	**47.1**	**.325**	**67.1**	**.189**	**1.7**	**.449 [.363,.559]**

Table 4: Stage C (adaptive attacker, 12-identity subset): DreamBooth-LoRA similarity (Sim), success rate at $\tau = 0.30$ (SR, %), and retention RR relative to the FSMG T0 gap, under clean-source mixing and diffusion purification of the gallery. T0 reference: 0.236 protected vs. 0.407 clean (SR 12.1% vs. 97.5%). The adaptation budget is per identity (400-step fine-tuning in all rows).

Condition	Sim	SR	RR	Adaptation cost
mix-20%	.300	49.2	.63	+1 clean source, 20% pool
mix-50%	.365	87.9	.25	+1 clean source, 50% pool
purify-50	.388	92.1	.11	+1 img2img per gallery image

Table 5: Cost ledger per identity (8-image gallery). Protection cost measured on an RTX 4090; funnel cost per image on the same machine. $\kappa = c_{\text{prot}}/(8 \cdot c_{\text{funnel}})$.

Operation	Side	Cost	κ
FSMG (surrogate + PGD)	defender	~500 GPU-s	—
ASPL-lite (2 rounds)	defender	~1000 GPU-s	—
T1 JPEG transcode	platform	5 ms/img	$\sim 10^4$
T2 browser render	platform	40 ms/img	$\sim 1.6\times 10^3$
T4 super-resolution	user/agent	1.0 s/img	~60
T5 diffusion regen.	attacker	3.5 s/img	~18

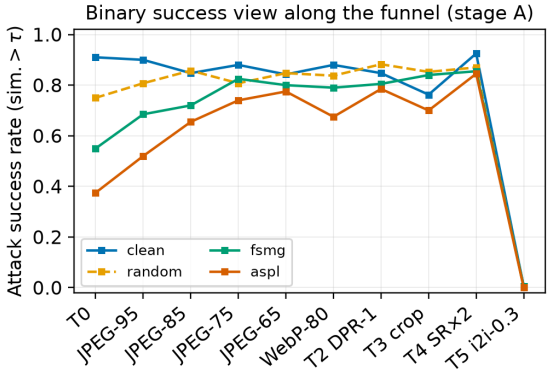


Figure 4: Binary success view: fraction of zero-shot generations whose best identity similarity clears $\tau = 0.30$, by tier. The clean baseline tracks what the funnel does to attack quality; the defenses' curves converge onto it as the funnel deepens.

5.4 The Cost Inversion

Table 5 completes the ledger. Forging an FSMG perturbation costs ~500 GPU-seconds per identity (250 surrogate-training steps plus 100 sign-PGD steps); ASPL-lite doubles it. Executing the tiers that defeat it costs 5 ms (JPEG transcode) to 3.5 s (diffusion regeneration) *per image* on the same class of hardware. Per identity, an 8-image gallery at the platform tiers costs the attacker under 0.4 CPU-seconds—a κ of three to four orders of magnitude in the defender's disfavor, shrinking to ~20× even at the most expensive tier (T5), which the attacker controls and can batch. The decisive asymmetry is not only in the ledger: the funnel's cheaper tiers are operated by platforms and users who have no stake in the attack, so the attacker pays only the attention cost of downloading.

5.5 Imperceptibility, the Binary View, and What the Attacker Sees

Protection quality is only meaningful relative to image quality. The measured perturbations sit at PSNR **28.1** dB (FSMG), **28.4** dB (ASPL-lite), and **29.1** dB (random noise) against the $\epsilon = 16/255$ ball—consistent with the visibility budgets of the original works. Under the binary view (Figure 4), the fraction of generations that clear the recognition threshold $\tau = 0.30$ follows the RR collapse; at $\tau = 0.25$ and 0.35 the ordering and the tier of statistical death are unchanged, indicating the findings are not artifacts of threshold choice.

Figure 5 grounds these statistics in what the attacker's pipeline actually consumes. The platform tiers (T1–T2) leave both the clean and the FSMG-protected gallery visually intact—which is exactly why they are dangerous: the funnel damages the *perturbation* without damaging the *photograph*. The costly tiers invert the picture: after T5 regeneration at strength 0.3, the clean identity still yields recognizable faces, while the FSMG-protected gallery is degraded past the point where the face detector inside the attack stack finds a face at all. This is protection by destruction, not by deception: the defender wins the cell, but by destroying the asset's value rather than by steering the attacker's model, and the honest framing matters for deployers—a social-media host that regenerates uploads at strength 0.3 would destroy face memorability for *everyone*, protected or not.

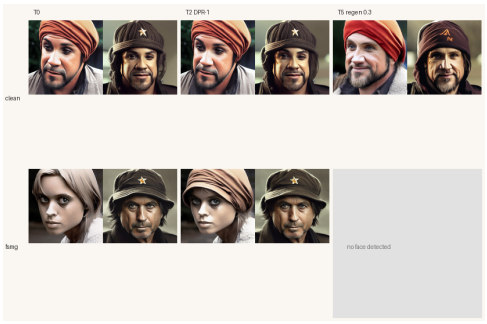


Figure 5: What the attacker actually receives, on one clean-protected and one FSMG-protected identity (two generations per cell, seed 0). Platform tiers leave both galleries visually intact; the adversarially expensive tiers do the defenses' work for them—T5 regeneration at strength 0.3 leaves the clean identity recognizable while the FSMG-protected gallery is degraded past face detectability, at which point the attack is not merely inaccurate but infeasible.

6 DISCUSSION: WHAT SURVIVES THE FUNNEL

6.1 Why Perturbation-Based Protection Is Structurally Fragile

The collapse we measure is not a collection of implementation artifacts; it follows from what protective perturbations are. A perturbation is a low-amplitude, typically high-frequency signal tuned to a specific computational path—the gradient of a fine-tuning loss through a particular denoiser. The funnel's platform tiers were never designed to preserve such signals: transcoders optimize rate–distortion, browsers resample for display, agents crop for relevance. Each operation is individually benign for semantic content and individually lethal for the perturbation's supporting structure, and their composition multiplies. Meanwhile the zero-shot attack family removes the very surface the perturbation poisons: IP-Adapter-FaceID reads an identity embedding off the funnel output and injects it through cross-attention, so there is no fine-tuning gradient to corrupt. A defense family whose mechanism is gradient poisoning and whose robustness story is adversarial training over hand-picked transformations does not, as currently formulated, model the pipeline it is deployed behind—and so cannot certify itself against it. Recent limits-focused analyses reach consonant conclusions from the laboratory side [2]; our contribution is to show that no laboratory attacker is needed—the commodity web is adversary enough.

6.2 A Reporting Standard: RR@T

We propose that protection papers report, alongside T0 numbers: (i) the collapse curve $\mathrm{RR}(T)$ over the public funnel tiers, (ii) the tier at which the Holm-corrected gap first becomes statistically indistinguishable from zero, and (iii) the cost ledger $(c_{\mathrm{prot}}, c_{\mathrm{funnel}}, \kappa)$. This costs a page and changes the field's incentives: a defense that is strong only at T0 can no longer headline a single number. The tiers are intentionally implementable with commodity libraries so that adoption is a scripting exercise, not a research project; our release ships the registry.

6.3 The Durable Fix: Consent Provenance, Not Stronger Noise

If F1–F3 hold, the marginal return on stronger perturbations is small: the funnel is free and adaptive tiers launder what survives. The alternative is to change what is being protected. A machine-readable consent manifest—`face-consent.txt` served beside the image, declaring identity, scope (train/generate/none), and revocation—converts face protection from a pixel-level arms race into a provenance problem that platforms can enforce at ingestion and scrapers can be required to honor. Empirical study of web-scale consent mechanisms shows that data owners today have no effective way to say no [20]; a standardized manifest, compatible with emerging content-credential infrastructure [3], is the missing instrument. This reframing has three advantages: it survives the funnel by construction (consent is metadata, not pixels); it composes with content credentials rather than fighting them; and it gives benign ecosystem actors—platforms, model trainers, inference providers—a technically checkable obligation, which perturbations never provide. We view ScrapeBench's role here as motivational: by quantifying exactly how little of today's pixel-level protection survives acquisition, the funnel makes the case for structural remedies concrete.

6.4 Limitations

Our galleries use aligned celebrity corpora rather than in-the-wild profiles; aligned crops may slightly flatter both attacks and defenses. The funnel tiers are faithful reconstructions, not captured traces from specific platforms; platform-internal pipelines vary and may be harsher or gentler than our composites. We test representative defenses from the open literature rather than every published variant, and our zero-shot attacker uses one adapter family; other identity adapters (e.g., InstantID-class [44]) may interact differently with T4–T5. Finally, ScrapeBench's adversary is resource-bounded and non-adaptive by design; a defense-aware red team with platform-grade infrastructure sits outside our threat model, and our bounds should be read as lower bounds on attacker capability.

7 CONCLUSION

We introduced ScrapeBench, an evaluation protocol that treats the web acquisition funnel—transcoding, browser and thumbnail rendering, agent capture, restoration, and diffusion-based regeneration—as the primary adversary of face anti-customization defenses. Measuring two published protections and two controls across two attack families under ten funnel tiers, we found that protections optimized for the laboratory's clean domain lose roughly half of their protection gap to a single JPEG-85 re-encode and nearly all of it by the browser tier, that zero-shot identity injection extracts comparable identity without touching the fine-tuning gradients that perturbation defenses poison, and that modest data-level adaptation—clean-source mixing and diffusion purification—closes the remaining gap. Because funnel transformations cost orders of magnitude less than protection generation and are selected by parties with no stake in the attacker's success, the arms race they anchor is structurally unwinnable. ScrapeBench's release—a tier

registry, replayable runners, and an RR-based reporting standard—is designed to make collapse curves, not clean-domain numbers, the field's currency, and to redirect research from stronger noise toward consent provenance and enforceable platform obligations.

REFERENCES

[1] Pu Cao, Feng Zhou, Qing Song, and Lu Yang. 2026. Controllable Generation with Text-to-Image Diffusion Models: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48, 4 (2026), 4771–4791. https://doi.org/10.1109/TPAMI.2025.3646548

[2] Zhen Chen, Yi Zhang, Xiangyu Yin, Chengxuan Qin, Xingyu Zhao, Xiaowei Huang, and Wenjie Ruan. 2025. Fragile by Design: On the Limits of Adversarial Defenses in Personalized Generation. arXiv:2511.10382. arXiv:2511.10382 http://arxiv.org/abs/2511.10382v1

[3] Coalition for Content Provenance and Authenticity. 2026. C2PA Technical Specification: Content Credentials. C2PA Technical Specification, https://c2pa.org/specifications/.

[4] Jiankang Deng, Jia Guo, Jing Yang, Niannan Xue, Irene Kotsia, and Stefanos Zafeiriou. 2022. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 10 (2022), 5962–5979. https://doi.org/10.1109/TPAMI.2021.3087709 arXiv:1801.07698

[5] Tanjim Rahaman Fardin, S M Zunaid Alam, Mahadi Hasan Fahim, and Md Faysal Mahfuz. 2026. MetaCloak-JPEG: JPEG-Robust Adversarial Perturbation for Preventing Unauthorized DreamBooth-Based Deepfake Generation. arXiv:2604.18537. arXiv:2604.18537 http://arxiv.org/abs/2604.18537v1

[6] Shaopeng Fu, Fengxiang He, Yang Liu, Li Shen, and Dacheng Tao. 2023. Robust Unlearnable Examples: Protecting Data Against Adversarial Learning. In *International Conference on Learning Representations (ICLR)*.

[7] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H. Bermano, Gal Chechik, and Daniel Cohen-Or. 2023. An Image is Worth One Word: Customizing Text-to-Image Generation using Textual Inversion. In *International Conference on Learning Representations (ICLR)*.

[8] Kefu Guo, Fengfan Zhou, Hefei Ling, Ping Li, and Hui Liu. 2024. Improving the JPEG-resistance of Adversarial Attacks on Face Recognition by Interpolation Smoothing. arXiv:2402.16586. arXiv:2402.16586 http://arxiv.org/abs/2402.16586v1

[9] Jiaming He, Jiashu Zhang, Guanyu Hou, Shuhan Ye, Hanwei Zhu, Yi Yu, and Xudong Jiang. 2026. ChronoLock: Protecting Videos from Unauthorized Text-to-Video Personalization. arXiv:2606.21146. arXiv:2606.21146 http://arxiv.org/abs/2606.21146v1

[10] Robert Hönig, Javier Rando, Nicholas Carlini, and Florian Tramèr. 2025. Adversarial Perturbations Cannot Reliably Protect Artists From Generative AI. In *International Conference on Learning Representations (ICLR)*.

[11] Hanxun Huang, Xingjun Ma, Sarah Monazam Erfani, James Bailey, and Yisen Wang. 2022. Unlearnable Examples: Making Personal Data Unexploitable. In *International Conference on Learning Representations (ICLR)*.

[12] Sangwon Jang, June Suk Choi, Jaehyeong Jo, Kimin Lee, and Sung Ju Hwang. 2025. Silent Branding Attack: Trigger-free Data Poisoning Attack on Text-to-Image Diffusion Models. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 8203–8212. https://doi.org/10.1109/CVPR52734.2025.00768

[13] Jaehwan Jeong, Sumin In, Sieun Kim, Hannie Shin, Jongheon Jeong, Sang Ho Yoon, Jaewook Chung, and Sangpil Kim. 2025. FaceShield: Defending Facial Image Against Deepfake Threats. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 10364–10374. https://doi.org/10.1109/ICCV51701.2025.00965

[14] Jun Jia, Hongyi Miao, Yingjie Zhou, Linhan Cao, Yanwei Jiang, Wangqiu Zhou, Dandan Zhu, Hua Yang, Wei Sun, Xiongkuo Min, and Guangtao Zhai. 2025. DLADiff: A Dual-Layer Defense Framework against Fine-Tuning and Zero-Shot Customization of Diffusion Models. arXiv:2511.19910. arXiv:2511.19910 http://arxiv.org/abs/2511.19910v1

[15] Wan Jiang, Yunfeng Diao, He Wang, Jianxin Sun, Meng Wang, and Richang Hong. 2023. Unlearnable Examples Give a False Sense of Security: Piercing through Unexploitable Data with Learnable Examples. In *Proceedings of the 31st ACM International Conference on Multimedia (ACM MM)*. ACM, 8910–8921. https://doi.org/10.1145/3581783.3611833

[16] Jungkon Kim, Cheolseung Jung, Jong-Min Choi, and Juseong Lee. 2026. Phantom: A Unified Face-Swap Deepfake Protection Framework with Latent and Spatial Constraints. In *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRF)*. IEEE, 847–856. https://doi.org/10.1109/CVPRF71909.2026.00086

[17] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. 2023. Multi-Concept Customization of Text-to-Image Diffusion. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 1931–1941. https://doi.org/10.1109/CVPR52729.2023.00192

[18] Thanh Van Le, Hao Phung, Thuan Hoang Nguyen, Quan Dao, Ngoc N. Tran, and Anh Tran. 2023. Anti-DreamBooth: Protecting users from personalized text-to-image synthesis. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2116–2127. https://doi.org/10.1109/ICCV51070.2023.00202

[19] Chanhui Lee, Donggyu Choi, Seunghyun Shin, Hae-Gon Jeon, and Jeany Son. 2026. Universal Image Immunization against Diffusion-based Image Editing via Semantic Injection. In *Lecture Notes in Computer Science*. Springer Nature Switzerland, 463–481. https://doi.org/10.1007/978-3-032-37167-6_27 arXiv:2602.14679

[20] Chung Peng Lee, Rachel Hong, Harry H. Jiang, Aster Plotnik, William Agnew, and Jamie Morgenstern. 2026. How Do Data Owners Say No? A Case Study of Data Consent Mechanisms in Web-Scraped Vision-Language AI Training Datasets. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 40. 38808–38816. https://doi.org/10.1609/aaai.v40i45.41225

[21] Ang Li, Yichuan Mo, Mingjie Li, and Yisen Wang. 2024. PID: Prompt-Independent Data Protection Against Latent Diffusion Models. arXiv:2406.15305. arXiv:2406.15305 http://arxiv.org/abs/2406.15305v1

[22] Jiahao Li, Yiqiang Chen, Yunbing Xing, Yang Gu, and Xiangyuan Lan. 2025. A Survey on Unlearnable Data. arXiv:2503.23536. arXiv:2503.23536 http://arxiv.org/abs/2503.23536v3

[23] Yixin Liu, Chenrui Fan, Yutong Dai, Xun Chen, Pan Zhou, and Lichao Sun. 2024. MetaCloak: Preventing Unauthorized Subject-Driven Text-to-Image Diffusion-Based Synthesis via Meta-Learning. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 24219–24228. https://doi.org/10.1109/CVPR52733.2024.02286

[24] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep Learning Face Attributes in the Wild. In *2015 IEEE International Conference on Computer Vision (ICCV)*. IEEE, 3730–3738. https://doi.org/10.1109/ICCV.2015.425

[25] Hengrui Lou, Zunlei Feng, Jinsong Geng, Erteng Liu, Lechao Cheng, Jie Lei, Jie Song, Mingli Song, and Yijun Bei. 2025. A Large-scale Universal Evaluation Benchmark For Face Forgery Detection. In *Proceedings of the 33rd ACM International Conference on Multimedia*. ACM, 12592–12599. https://doi.org/10.1145/3746027.3758194

[26] Takuto Onikubo and Yusuke Matsui. 2024. High-Frequency Anti-DreamBooth: Robust Defense against Personalized Image Synthesis. arXiv:2409.08167. arXiv:2409.08167 http://arxiv.org/abs/2409.08167v3

[27] Tarik Can Ozden, Ozgur Kara, Oguzhan Akcin, Kerem Zaman, Shashank Srivastava, Sandeep P. Chinchali, and James M. Rehg. 2024. DiffVax: Optimization-Free Image Immunization Against Diffusion-Based Editing. arXiv:2411.17957. arXiv:2411.17957 http://arxiv.org/abs/2411.17957v2

[28] Gan Pei, Jiangning Zhang, Menghan Hu, Zhenyu Zhang, Chengjie Wang, Yunsheng Wu, Guangtao Zhai, Jian Yang, and Dacheng Tao. 2026. Deepfake Generation and Detection: A Benchmark and Survey. *Comput. Surveys* 58, 11 (2026), 1–41. https://doi.org/10.1145/3801962

[29] Sen Peng, Mingyue Wang, Jianfei He, Jijia Yang, and Xiaohua Jia. 2025. CAT: Contrastive Adversarial Training for Evaluating the Robustness of Protective Perturbations in Latent Diffusion Models. arXiv:2502.07225. arXiv:2502.07225 http://arxiv.org/abs/2502.07225v2

[30] Tianrui Qin, Xitong Gao, Juanjuan Zhao, Kejiang Ye, and Cheng-Zhong Xu. 2024. Learning the Unlearnable: Adversarial Augmentations Suppress Unlearnable Example Attacks. In *International Conference on Learning Representations (ICLR)*.

[31] Chunlin Qiu, Ang Li, Tianxiao Huang, Ruilin Gan, Yunjie Ge, Shenyi Zhang, Huayi Duan, Lingchen Zhao, Chao Shen, and Qian Wang. 2026. VOID: Defeating Unauthorized Mimicry in Latent Diffusion Models. arXiv:2606.12263. arXiv:2606.12263 http://arxiv.org/abs/2606.12263v2

[32] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 10674–10685. https://doi.org/10.1109/cvpr52688.2022.01042

[33] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. 2023. DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 22500–22510. https://doi.org/10.1109/CVPR52729.2023.02155

[34] Hadi Salman, Alaa Khaddaj, Guillaume Leclerc, Andrew Ilyas, and Aleksander Madry. 2023. Raising the Cost of Malicious AI-Powered Image Editing. In *2023 IEEE Symposium on Security and Privacy (S&P)*.

[35] Pedro Sandoval-Segura, Jonas Geiping, and Tom Goldstein. 2023. JPEG Compressed Images Can Bypass Protections Against AI Editing. arXiv:2304.02234. arXiv:2304.02234 http://arxiv.org/abs/2304.02234v2

[36] Pedro Sandoval-Segura, Vasu Singla, Jonas Geiping, Micah Goldblum, and Tom Goldstein. 2023. What Can We Learn from Unlearnable Datasets? arXiv:2305.19254. arXiv:2305.19254 http://arxiv.org/abs/2305.19254v3

[37] Pedro Sandoval-Segura, Vasu Singla, Jonas Geiping, Micah Goldblum, Tom Goldstein, and David W. Jacobs. 2022. Autoregressive Perturbations for Data Poisoning. In *Advances in Neural Information Processing Systems (NeurIPS)*. Neural Information Processing Systems Foundation, 27374–27386. https://doi.org/10.52202/068431-1985

[38] Shawn Shan, Jenna Cryan, Emily Wenger, Haitao Zheng, Rana Hanocka, and Ben Y. Zhao. 2023. Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models. In *32nd USENIX Security Symposium (USENIX Security 23)*.
[39] Shawn Shan, Emily Wenger, Jiayun Zhang, Huiying Li, Haitao Zheng, and Ben Y. Zhao. 2020. Fawkes: Protecting Privacy against Unauthorized Deep Learning Models. USENIX Security Symposium 2020. arXiv:2002.08327 http://arxiv.org/abs/2002.08327v2
[40] Yiren Song, Pei Yang, Hai Ci, and Mike Zheng Shou. 2024. IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation. arXiv:2412.11638. arXiv:2412.11638 http://arxiv.org/abs/2412.11638v2
[41] Matteo Trippodo, Federico Becattini, and Lorenzo Seidenari. 2025. Immunizing Images from Text to Image Editing via Adversarial Cross-Attention. arXiv:2509.10359. arXiv:2509.10359 http://arxiv.org/abs/2509.10359v1
[42] Derui Wang, Minhui Xue, Bo Li, Seyit Camtepe, Liming Zhu, and Derui Wang. 2025. Provably Unlearnable Data Examples. In *Proceedings of the 2025 Network and Distributed System Security Symposium (NDSS)*. Internet Society. https://doi.org/10.14722/ndss.2025.240886
[43] Guanyu Wang, Kailong Wang, Yihao Huang, Mingyi Zhou, Geguang Pu, and Li Li. 2026. Privacy Protection Against Personalized Text-to-Image Synthesis via Cross-image Consistency Constraints. arXiv:2504.12747. In *Proceedings of the 2026 International Conference on Multimedia Retrieval (ICMR)*. ACM, 2003–2011. https://doi.org/10.1145/3805622.3810856 arXiv:2504.12747
[44] Qixun Wang, Xu Bai, Haofan Wang, Zekui Qin, Anthony Chen, Huaxia Li, Xu Tang, and Yao Hu. 2024. InstantID: Zero-shot Identity-Preserving Generation in Seconds. arXiv:2401.07519. arXiv:2401.07519 http://arxiv.org/abs/2401.07519v2
[45] Songping Wang, Yueming Lyu, Shiqi Liu, Ning Li, Tong Tong, Hao Sun, and Caifeng Shan. 2025. Anti-Aesthetics: Protecting Facial Privacy against Customized Text-to-Image Synthesis. arXiv:2504.12129. arXiv:2504.12129 http://arxiv.org/abs/2504.12129v2
[46] Songping Wang, Yueming Lyu, Shiqi Liu, Chen Zhao, Ziyuan Chen, Ning Li, Jing Dong, and Caifeng Shan. 2026. Hierarchical Anti-Aesthetics: Protecting Facial Privacy against Customized Diffusion Models. arXiv:2607.02038. arXiv:2607.02038 http://arxiv.org/abs/2607.02038v2
[47] Tao Wang, Yushu Zhang, Xiangli Xiao, Kun Xu, Lin Yuan, Wenying Wen, and Yuming Fang. 2025. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. arXiv:2509.11249. arXiv:2509.11249 http://arxiv.org/abs/2509.11249v1
[48] Qiangqiang Wu, Yi Yu, Chenqi Kong, Ziquan Liu, Jia Wan, Haoliang Li, Alex C. Kot, and Antoni B. Chan. 2025. Temporal Unlearnable Examples: Preventing Personal Video Data from Unauthorized Exploitation by Object Tracking. In *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 11110–11121. https://doi.org/10.1109/ICCV51701.2025.01034
[49] Stanley Wu, Ronik Bhaskar, Anna Yoo Jeong Ha, Shawn Shan, Haitao Zheng, and Ben Y. Zhao. 2025. On the Feasibility of Poisoning Text-to-Image AI Models via Adversarial Mislabeling. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2848–2862. https://doi.org/10.1145/3719027.3744845
[50] Zhihao Wu, Yushi Cheng, Tianyang Sun, Xiaoyu Ji, and Wenyuan Xu. 2025. MYOPIA: Protecting Face Privacy from Malicious Personalized Text-to-Image Synthesis via Unlearnable Examples. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 39. 905–913. https://doi.org/10.1609/aaai.v39i1.32075
[51] Wenkui Yang, Jie Cao, Junxian Duan, and Ran He. 2025. Towards Robust Defense against Customization via Protective Perturbation Resistant to Diffusion-based Purification. arXiv:2509.13922. arXiv:2509.13922 http://arxiv.org/abs/2509.13922v2
[52] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. arXiv:2308.06721. arXiv:2308.06721 http://arxiv.org/abs/2308.06721v1
[53] Tong Zhang, Ru Zhang, and Jianyi Liu. 2026. From Concept Erasure to Style Purification: Contrastive Eigenbases for Artist Style Protection. arXiv:2602.08059. arXiv:2602.08059 http://arxiv.org/abs/2602.08059v2
[54] Boyang Zheng, Chumeng Liang, and Xiaoyu Wu. 2023. Targeted Attack Improves Protection against Unauthorized Diffusion Customization. arXiv:2310.04687. arXiv:2310.04687 http://arxiv.org/abs/2310.04687v5

A APPENDIX

A.1 Reproducibility

The complete harness (`scrapebench`) is released under an open license with: (i) the tier registry implementing T0–T6 as pure image functions with deterministic fallbacks; (ii) per-stage runners (`protection_run`, `stage_a`, `stage_b`, `stage_c`) writing append-only CSVs keyed by (stage, identity, defense, tier, attack) and resuming deterministically; (iii) the report generator producing collapse curves, retention tables, paired Wilcoxon tests with Holm correction, Cliff's δ, and bootstrap CIs; (iv) a cost ledger emitted per cell. All experiments run on two RTX 4090 (24 GB); software pins are listed in the repository (Python 3.10, torch 2.x CUDA, diffusers 0.31, transformers 4.44, peft 0.13, insightface 0.7). Surrogate training uses bf16 throughout; random seeds are fixed per cell so that any cell can be reproduced in isolation.

A.2 Ethics Statement

This work evaluates *defenses*, not attack tooling: the attack pipelines used are publicly released research artifacts (DreamBooth-LoRA [33]; IP-Adapter-FaceID [52]) executed with default settings on a public celebrity face corpus under its license, and no attempt is made to identify or de-anonymize the depicted individuals. Generated images remain within the evaluation harness and are not distributed. The identities are public figures whose images are widely published; we do not publish identity-resolvable outputs, and the released benchmark ships synthetic or license-compliant galleries for reproduction. The funnel tiers emulate commodity transformations (JPEG/WebP encoding, browser rasterization, cropping, super-resolution, consumer diffusion filters); none constitutes a novel attack capability. We hope the measured collapse curves inform platform policy: our results argue for consent manifests and ingestion-time obligations rather than stronger pixel noise, which is precisely the non-adversarial, harm-reducing direction. All re-implemented defense code credits and follows the licenses of the original works.

A.3 Additional Results

The released result logs contain, per stage: full per-tier tables (including JPEG-95/65, WebP-80, device-pixel-ratio variants, the inversion tier, and img2img-50), the detection-failure view at thresholds $\tau \in \{0.25, 0.30, 0.35\}$, per-identity collapse curves with bootstrap intervals, quality metrics (PSNR/LPIPS) of every protection under every tier, per-cell wall-clock timings, and the complete Wilcoxon–Holm test table. The preview artefacts accompanying this submission (`runs/preview/`, `figures/`) are regenerable with a single `report` invocation from any results CSV in the same schema.

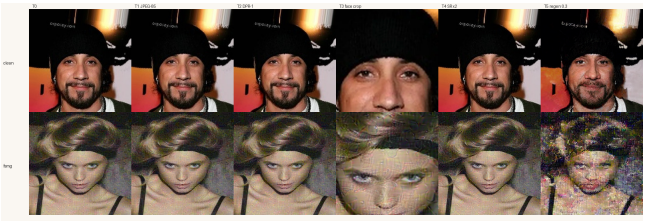


Figure 6: The six Stage-A funnel tiers applied to one clean (top) and one FSMG-protected (bottom) gallery image. Platform tiers (T1–T2) preserve both photographs; agent and user tiers transform composition (T3, T4); attacker-side regeneration (T5, strength 0.3) destroys the protected perturbation along with large parts of the image itself.

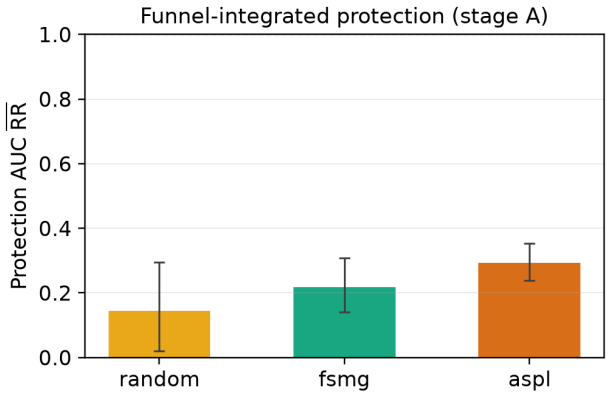


Figure 7: Funnel-integrated protection $\overline{\mathrm{RR}}$ for the Stage-A defenses (95% bootstrap CIs over cells). The integrated view compresses each curve of Figure 2 into a single deployment-relevant number.

Figure 6 shows the tiers themselves operating on real gallery images, and Figure 7 the integrated protection view for Stage A.